MIL-P-52981
1 August 1980
SUPERSEDING
MIL-P-20664C
9 February 1977

MILITARY SPECIFICATION

PLANIMETERS, POLAR, COMPENSATING, WITH CASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers compensating polar planimeters which are utilized for measuring areas and calculating volume from a plane figure by tracing or optically observing its boundary lines.

1.2 Classification.

Type I - Tracer arm with tracer needle
Type II - Tracer arm with optical tracer.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: *DRDME-DS*, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:45Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specification and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

3. REQUIREMENTS

3.1 Description. The planimeter shall be of the compensating polar type. The planimeter shall consist of a tracer arm, which carries the tracing point, a measuring mechanism with not less than six place electronic digital processor with prepared program cards having optional English or metric readings in both visual and printed record, electronic zero setting and accumulative measuring, a linear measuring probe, a point counter, a remote foot switch, not less than six feet of extension cord for the planimeter, a test ruler and a pole arm attached to a pole around which the instrument revolves. The equipment shall operate on 110 volts 60 cycle current.

3.2 First article (first produced) planimeter. The contractor shall furnish one or more planimeters as specified (see 6.2), for examination and testing within the time frame specified (see 6.2), to prove that his production methods and choice of design detail will produce (planimeters) that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4).

3.3 Material. Material shall be corrosion resistant as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:45Z
Check the source to verify that this is the current version before use.

3.4 Tracer arm. The tracer arm shall be provided with a tracer needle (type I) or optical tracer (type II) as specified (see 6.2). The arm shall be graduated for an effective length of not less than 6 inches to enable setting the recording carriage vernier to hundredths at any scale within the range of the instrument. The tracer needle shall have an adjustable point. The tracer needle or optical tracer shall be provided with a fingertip guider positioned at the outer end of the tracer arm.

3.5 Arms. Adjustable tracer arm shall have a length of 10 inches (25.4 centimeters). Adjustable pole arm shall have a maximum length of 13.5 inches (35 centimeters). Error shall not be more than 0.2 percent. The error in the area of maximum resolution shall not be more than 0.009 inches squared (0.05 centimeters squared); linear error shall not be more than 0.0002 inches (0.006 centimeters). Measuring range shall be not less than 12 x 28 inches (30 x 71 centimeters) (pole outside). Linear measuring range shall be unlimited.

3.6 Recording mechanism. The measuring wheel of electronic planimeters shall be attached to an electro-optical shaft encoder. The measuring encoder shall generate exactly 1000 pulses for each revolution of the measuring wheel. The measuring wheel shall be made from SAE (Society of Automotive Engineers) 1075 spring steel, hardened to Rockwell C55-65 and lacquered against corrosion.

3.7 Graduations. The graduations on the tracer arm and pole arm shall be machine divided and shall be spaced no greater than 1.0 millimeter (mm) when the metric system is used, or 0.025 inch when the decimal inch system is used, shall be straight, uniform, parallel, permanently marked, and free from breaks or ragged edges, and shall be clear and distinct so as to be quickly and easily read. Graduation lines of equal lengths shall indicate aliquot parts of the scale. Lines indicating subdivisions of larger parts shall be shorter than the lines indicating major divisions so that the difference is distinguishable at a glance. Vernier scales shall read to one-tenth of corresponding least scale divisions. The error between any two graduation marks shall not exceed 0.002 inch.

3.8 Area coverage. The planimeter shall measure the area within a circle having a 16-inch radius when the pole is within the circle and shall measure the area of a circle 12 inches in diameter when the pole is outside of the circle.

3.9 Testing rule. A metal testing rule shall be furnished with each planimeter. The rule shall have an engraved indexing mark at one end and a 1/16-inch positioning needle projecting from the side opposite the indexing mark at the other end. Two holes shall be drilled to snugly fit the tracing point of the planimeter on a line connecting the index mark and the positioning needle. The holes on the testing rule shall be at positions which will give 12.56 square inches of area. The areas of the circles enscribed by rotating the drilled hole about the needle point as a center shall be engraved on the rule beside their corresponding hole, and the numerical value shall be within plus or minus 1 percent of the area computed by using the spacing between the positioning needle and the center of the drilled hole as the radius of a circle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:45Z
Check the source to verify that this is the current version before use.

3.10 Accuracy. The instrument shall check the areas engraved on the testir rule indicated on the graphs within 1.0 percent.

3.11 Case. The planimeter shall be furnished in a fitted case which shall lined with a material which is fungus resistant or treated to be fungus resistant. The case shall have a hinged cover and a fastener. The material used to fabricate the case shall be fungus resistant or treated to be fungus resistant.

3.12 Identification marking. The planimeter, processing system and case shall be identified in accordance with MIL-STD-130.

3.13 Workmanship. All parts of the planimeter shall be clean and free from sand, dirt, fins, spruces, scale, flux, or other harmful extraneous material. All edges shall be rounded and chamfered.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specififed in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any the inspections set forth in the specification where such inspections are deem necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirments specified herein are classified as follows:

(a) First produced inspections (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced inspection.

4.3.1 Examinations. The first produced planimeters shall be examined as specified in 4.5.1. The presense of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced planimeters shall be tested as specified i 4.5.2. Failure of any test shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:45Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 2.5 percent defective.

4.4.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.5.2.1 through 4.5.2.4.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The planimeters shall be examined as specified herein for the following major defects:

101. Materials not as specified.
102. Recording mechanism, when unlocked, does not slide freely on the tracer arm.
103. Clamping and slow-motion screws bind or do not turn freely.
104. Measuring and support wheels do not turn freely.
105. Graduations not as specified.
106. Pole arm not adjustable within specified limits.
107. Identification marking not as specified.
108. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Area coverage. Set up the planimeter at maximum scale-setting range and move the tracing point so as to describe the maximum circle. Inability of the tracing point to describe a circle having a minimum radius of 16 inches shall constitute failure of this test.

4.5.2.2 Testing-rule accuracy. Check the area value engraved on the rule of the planimeter by measuring the spacing between the needle point and the center of the drilled hole to within 0.01 inch for the inch area and 0.05 centimeter for the metric area and compute the area of the circle which would result from using this spacing as a radius. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.3 Area-measuring accuracy. Check the planimeter for area-measuring accuracy using the following area designations engraved on the testing rule:

(a) The 10-square-inch value. Set the mechanism at zero, insert the needle point of the test rule into the drafting surface, and place the tracing arm point of the planimeter into the hole in the testing rule having 10 square inches engraved beside it. Mark on the drafting surface the beginning as shown by the indexing mark on the rule and then rotate the rule about the impressed point as a center; stop at

Source: https://assist.dla.mil -- Downloaded: 20261O-22T21:45Z
Check the source to verify that this is the current version before use.

the beginning mark after completing 360 degrees. Record the total square inches for one revolution of the test rule. Rotate the rule two more times around the impressed point, each time recording the number of gains for one revolution of the test rule. Compute the average of the three revolutions. An average value not within plus or minus 1.0 percent of 10 square inches shall constitute failure of this test.

(b) The 100-square-centimeter value. Repeat the foregoing procedure (a) but use the hole on the testing rule for centimeters and convert the average value to square centimeters. An average value not within 1.0 percent of 100 square centimeters shall constitute failure of this test.

4.5.2.4 Volume estimating. Construct a drawing one meter by one meter with estimated depth as one meter. By using the planimeter with Quality Processing System, estimate the volume. The estimated volume not being within 1 percent of the calculated volume shall be cause for rejection.

4.5.2.4.1 Counting points. Establish a group of 60 random points. By use of the Quality Processing System count the points established on the paper. The number of points counted not being within 1 percent of the number of points established shall be cause for rejection.

4.5.2.4.2 Length of line. Establish a line of one meter in length. Measure the line with the planimeter. The planimeter measurement being more than 1 percent different from the established line shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

109. Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
110. Stapping not as specified.
111. Marking illegible, incomplete, incorrect, or missing.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:45Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, or commercial, as specifed (see 6.2).

5.1.1 Level A. Each complete planimeter, in its transit case, and each complete auto-scale, in its transit case, shall be preserved individually in accordance with MIL-P-116, method IA-8.

5.1.1.1 Consolidation. One each planimeter and one each auto-scaler, preserved as specified in 5.1.1, shall be placed together in a close fitting fiberboard box conforming to PPP-B-636, type W6c, style optional.

5.1.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. Planimeters and components, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. Strapping shall conform to QQ-S-781, class 1, type I or type IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A. Total weight of container and contents shall not exceed 250 lbs.

5.2.2 Level B. Packing shall be as specified for level A, except that boxes shall be domestic type and strapping shall be finish A.

5.2.3 Commercial. Commercial packing shall be in accordance with MIL-STD-1188.

5.3 Marking. Marking for military levels of protection shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The planimeter is intended to determine the area, length of line, and calculated volume of figures with irregular or curved boundaries.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of specification.
(b) Type tracer required (see 1.2 and 3.4).
(c) Time frame required for submission of first article and number of planimeters required (see 3.2).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(e) When other than finish B strapping is required (see 5.2.1).

Source: https://assist.dla.mil -- Downloaded: 20260-22T21:45Z
Check the source to verify that this is the current version before use.

6.3 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each planimeter.

6.4 First produced planimeter. Any changes or deviations of production planimeters from the approved first produced planimeter during production will be subject to the approval of the contracting officer. Approval of the first produced planimeter will not relieve the contractor of his obligation to furnish planimeters conforming to this specification.

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3)

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of planimeters.

Custodian:
Army - ME
Air Force - 99

User activity:
Army - ER

Review activity:
DLA - GS

Preparing activity:
Army - ME

Project 6675-A232

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:45Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-52981 Planimeters, Polar, Compensating, with Case

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:45Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
USA Mobility Equipment R&D Command
ATTN: DRDME-DE/DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:45Z
Check the source to verify that this is the current version before use.